

ShelbyFirst Table Chart

Python source rendered from shelbyfirst_table_chart.py.

Approximate line count: 232

```
#!/usr/bin/env python3
"""
ShelbyFirst: Outcomes Data Table (standalone)
Run: python shelbyfirst_table_chart.py
Output: src/data/processed/shelbyfirst_table_chart.png
"""

import matplotlib
matplotlib.use('Agg')
import matplotlib.pyplot as plt
from matplotlib.patches import Rectangle
import numpy as np
import pandas as pd
import os
from sklearn.linear_model import Ridge
from sklearn.preprocessing import StandardScaler
from sklearn.pipeline import Pipeline

# — Palette —————
NAV = '#1b2a6b'
TBL = '#2e4296'
WHT = '#ffffff'
EVN = '#eef1fa'
GRY = '#dee3ef'
RED = '#c0392b'
GRN = '#1a7a35'
BLU = '#1a4db5'
DRK = '#1a1a2e'
SUB = '#8898cc'

# — Model-driven projections —————
_dir = os.path.dirname(os.path.abspath(__file__))
_data = os.path.join(_dir, 'data', 'processed')

_panel = pd.read_parquet(os.path.join(_data, 'enrichment_tract_year_panel.parquet'))
_shelby = _panel[_panel['geoid'].astype(str).str.startswith('47157')].copy()
_shelby['income_k'] = _shelby['median_household_income'] / 1000

_YEARS = [2024, 2025, 2026, 2027]
UPTAKE = [0.0, 0.15, 0.35, 0.60]

_PEER = {
    'labor_force': 66.9, 'poverty': 5.5, 'income': 119.0,
    'insured': 94.0, 'no_veh': 4.2, 'dental': 32.1,
    'obesity': 32.5, 'diabetes': 11.3,
}

EFFECT = {
    'labor_force': 0.22,
```

```

    'poverty':      0.15,
    'income':       0.10,
    'insured':      0.40,
    'no_veh':       0.20,
    'dental':       0.35,
    'obesity':      0.22,
    'diabetes':     0.18,
}

def _ridge_trend(df, col, scale=1.0):
    d = df[['year', col]].dropna()
    pipe = Pipeline([('sc', StandardScaler()), ('r', Ridge(alpha=1.0))])
    pipe.fit(d[['year']].values, d[col].values)
    return (pipe.predict(np.array([[yr] for yr in _YEARS])) * scale).tolist()

def _gap_boost(baseline, peer, pillar, direction):
    gap = abs(peer - baseline)
    eff = EFFECT[pillar]
    return [baseline + direction * gap * eff * u for u in UPTAKE]

def _econ(df, col, scale, peer, pillar, direction):
    trend = _ridge_trend(df, col, scale)
    gap = abs(peer - trend[0])
    eff = EFFECT[pillar]
    result = [trend[0]]
    for i in range(1, 4):
        result.append(trend[i] + direction * gap * eff * UPTAKE[i])
    return result

_lfpr_v = _econ(_shelby, 'lfpr_16p', 100, _PEER['labor_force'], 'labor_force', 1)
_pov_v = _econ(_shelby, 'poverty_rate', 100, _PEER['poverty'], 'poverty', -1)
_inc_v = _econ(_shelby, 'income_k', 1, _PEER['income'], 'income', 1)

_cs = pd.read_parquet(os.path.join(_data, 'cluster_shortlist_with_healthcare.parquet'))
_r77 = _cs[_cs['rank'] == 11]
_ins0 = (1 - float(_r77.iloc[0]['uninsured_rate'])) * 100 \
    if len(_r77) and not pd.isna(_r77.iloc[0]['uninsured_rate']) else 82.0

_ins_v = _gap_boost(_ins0, _PEER['insured'], 'insured', 1)
_noveh_v = _gap_boost(13.8, _PEER['no_veh'], 'no_veh', -1)
_dental_v = _gap_boost(55.6, _PEER['dental'], 'dental', -1)
_obes_v = _gap_boost(45.2, _PEER['obesity'], 'obesity', -1)
_diab_v = _gap_boost(21.6, _PEER['diabetes'], 'diabetes', -1)

_IGS = [35.8, 36.8, 38.4, 40.5]

METRICS = [
    ('IGS Score', 1, _IGS, 55.7,
     '#4545c0', '', '{:.1f}'),
    ('Labor Force %', 1, _lfpr_v, _PEER['labor_force'],
     '#1a8c5a', '', '{:.1f}%'),
    ('Insured %', 1, _ins_v, _PEER['insured'],
     '#c9a000', '', '{:.1f}%'),
    ('Med. Income', 1, _inc_v, _PEER['income'],
     '#e06000', '', '${:.1f}k'),

```

```

    ('Poverty %', -1, _pov_v, _PEER['poverty'],
     '#8b1515', '', '{:.1f}%'),
    ('No-Veh HH %', -1, _noveh_v, _PEER['no_veh'],
     '#cc1515', '', '{:.1f}%'),
    ('Dental Gap %', -1, _dental_v, _PEER['dental'],
     '#1a8cd4', '', '{:.1f}%'),
    ('Obesity %', -1, _obes_v, _PEER['obesity'],
     '#8bc820', '', '{:.1f}%'),
    ('Diabetes %', -1, _diab_v, _PEER['diabetes'],
     '#30a040', '', '{:.1f}%'),
]

# — Figure
fig = plt.figure(figsize=(12, 9), dpi=130, facecolor=NAV)

# — Header
fig.text(0.030, 0.965,
        'ShelbyFirst: Projected Outcomes',
        color=WHT, fontsize=22, fontweight='bold', va='top')
fig.text(0.030, 0.922,
        'ShelbyFirst gap-closing model (Move · Nourish · Connect) · '
        'Uptake: 15% Pilot → 35% Expand → 60% Scale · 96 tracts · pop. 273K',
        color=SUB, fontsize=10.5, va='top')

# — Table
ax_t = fig.add_axes([0.030, 0.115, 0.945, 0.775])
ax_t.set_xlim(0, 1)
ax_t.set_ylim(0, 1)
ax_t.axis('off')

COL_W = [0.280, 0.112, 0.108, 0.108, 0.108, 0.200]
COL_X = list(np.cumsum([0.0] + COL_W[:-1]))
COL_H = ['Metric', 'Now (2024)', 'Year 1 2025', 'Year 2 2026', 'Year 3 2027', 'High IGS Shelby County']

ROW_H_HDR = 0.095
ROW_H_SEC = 0.070
ROW_H_DATA = 0.155

TABLE_ROWS = [
    ('hdr', None),
    ('sec', '— IGS SCORE · Planning target – rises as a consequence of ShelbyFirst programs —'),
    ('data', METRICS[0]),
    ('sec', '— HEALTH METRICS · ShelbyFirst Move (rides → dental access) / Nourish (garden → obesity, diabetes)'),
    ('data', METRICS[6]),
    ('data', METRICS[7]),
    ('data', METRICS[8]),
]

y = 0.99
data_idx = 0

for rtype, rcontent in TABLE_ROWS:
    rh = {'hdr': ROW_H_HDR, 'sec': ROW_H_SEC, 'data': ROW_H_DATA}[rtype]

    if rtype == 'hdr':
        for i, (cx, cw, ch) in enumerate(zip(COL_X, COL_W, COL_H)):
            ax_t.add_patch(Rectangle((cx, y - rh), cw, rh,
                                     facecolor=NAV, edgecolor='#3a4ea0',
                                     linewidth=0.6, zorder=2))

```

```

        xt = cx + 0.015 if i == 0 else cx + cw / 2
        fs = 10.5 if i == len(COL_H) - 1 else 13
        ax_t.text(xt, y - rh / 2, ch,
                  ha='left' if i == 0 else 'center', va='center',
                  fontsize=fs, color=WHT, fontweight='bold')

    elif rtype == 'sec':
        ax_t.add_patch(Rectangle((0, y - rh), 1.0, rh,
                                facecolor=TBL, edgecolor='none', zorder=2))
        ax_t.text(0.015, y - rh / 2, rcontent,
                  ha='left', va='center', fontsize=9,
                  color='#aac8ff', style='italic')

    else:
        tbl_lbl, direction, vals, peer, color, _, fmt = rcontent
        bg = EVN if data_idx % 2 == 1 else WHT
        data_idx += 1

        ax_t.add_patch(Rectangle((0, y - rh), 1.0, rh,
                                facecolor=bg, edgecolor=GRY,
                                linewidth=0.4, zorder=1))

        # Left color accent bar
        ax_t.add_patch(Rectangle((0, y - rh), 0.006, rh,
                                facecolor=color, edgecolor='none', zorder=3))

        for cx in COL_X[1:]:
            ax_t.plot([cx, cx], [y - rh, y], color=GRY, linewidth=0.5, zorder=2)

        ax_t.text(COL_X[0] + 0.018, y - rh / 2, tbl_lbl,
                  ha='left', va='center', fontsize=14, color=DRK, fontweight='semibold')

        ax_t.text(COL_X[1] + COL_W[1] / 2, y - rh / 2, fmt.format(vals[0]),
                  ha='center', va='center', fontsize=14, color=RED, fontweight='bold')

        ax_t.text(COL_X[2] + COL_W[2] / 2, y - rh / 2, fmt.format(vals[1]),
                  ha='center', va='center', fontsize=14, color=DRK)

        ax_t.text(COL_X[3] + COL_W[3] / 2, y - rh / 2, fmt.format(vals[2]),
                  ha='center', va='center', fontsize=14, color=DRK)

        ax_t.text(COL_X[4] + COL_W[4] / 2, y - rh / 2, fmt.format(vals[3]),
                  ha='center', va='center', fontsize=15, color=GRN, fontweight='bold')

        ax_t.text(COL_X[5] + COL_W[5] / 2, y - rh / 2, fmt.format(peer),
                  ha='center', va='center', fontsize=14, color=BLU, fontweight='bold')

    y -= rh

# — Footer —————
fig.text(
    0.030, 0.012,
    'Model: improvement = (peer gap) × program effectiveness × uptake . '
    'Econ metrics add Ridge year-trend as counterfactual baseline . '
    'Effectiveness grounded in Move (rides), Nourish (garden), Connect (app/referrals) pillars . PDF slides 6-8',
    color=SUB, fontsize=8, va='bottom',
)

# — Save —————

```

ShelbyFirst Table Chart

```
out = os.path.join(_dir, 'data', 'processed', 'shelbyfirst_table_chart.png')
os.makedirs(os.path.dirname(out), exist_ok=True)
plt.savefig(out, dpi=130, bbox_inches='tight', facecolor=NAV, pad_inches=0.2)
print(f'Saved: {out}')
plt.close()
```